Tech Dessert #1: The Amateur Service and Your Frequency Privileges

Take-home reference -- what amateur radio is for, who runs it, and exactly where and how a Technician may transmit.

Groups T1A + T1B · 23 pool questions · Self-study handout · Practice exams at eham.net/exams

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This Dessert is a reference sheet – there is nothing to figure out, just facts to learn. Read the tables, then drill the question bank on the back until the answers feel automatic. These are some of the easiest points on the whole exam, so lock them in early and build your confidence.

What Amateur Radio Is, and Who Runs It

Topic	What to know	Q
What the Amateur Service is for	Advancing skills in the technical and communication phases of the radio art.	T1A01
Who makes and enforces the rules	The **FCC** – the Federal Communications Commission.	T1A02
How you learn you passed and got a call sign	An **email from the FCC** with a link to download your license grant.	T1A04
What actually proves you are licensed	Your license **appears in the FCC ULS database**.	T1A05
When to use a phonetic alphabet (Alpha, Bravo, Charlie…)	It is **encouraged** whenever you use phone (voice).	T1A03
What is always prohibited	**Willful or malicious interference** with other stations.	T1A11

Coordinators, Beacons, and Key Definitions

Topic	What to know	Q
Who recommends repeater transmit/receive channels	A **Volunteer Frequency Coordinator** recognized by local amateurs.	T1A08
Who selects that Frequency Coordinator	The local amateurs whose own stations are eligible to be repeater or auxiliary stations.	T1A09
Where HF propagation beacons are found	On 10 meters, between **28.200 and 28.300 MHz**.	T1A06
What “space station” means in FCC rules	An amateur station located **more than 50 km above the Earth**.	T1A07
Extra requirement to be control operator of a RACES station	**Certification of current enrollment** by a civil defense organization.	T1A10

Where Technicians Can Transmit

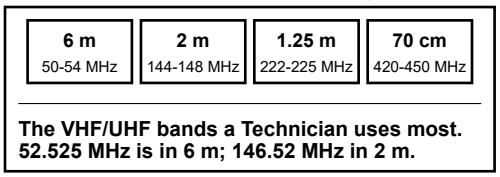
Topic	What to know	Q
A frequency in the 6-meter band	**52.525 MHz**.	T1B03
The band that includes 146.52 MHz	**2 meters**.	T1B04
The only HF band with Technician phone privileges	**10 meters** – and only 10 meters.	T1B06
The 10-meter segment for Technician phone	**28.300 to 28.500 MHz**.	T1B01
VHF/UHF segments limited to CW only	**50.0-50.1 MHz** and **144.0-144.1 MHz**.	T1B07

Modes, Power, and Good Practice

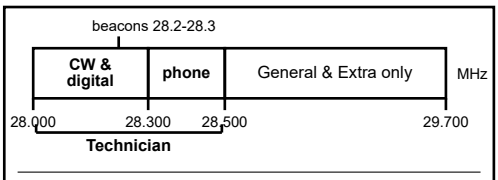
Topic	What to know	Q
Bands where Technicians may use digital modes like FT8	**10, 6, and 2 meters** – all of them.	T1B05
License needed to contact the ISS on VHF	**Technician or higher** – your license already qualifies.	T1B02

Where SSB phone is allowed above 50 MHz	In **at least some segment of every one** of those bands.	T1B10
What “secondary” status on a band means	You may find non-amateur stations there, and you must **avoid interfering** with them.	T1B08
Why not to transmit right at a band edge	So your signal’s sidebands stay **inside** the band (also allows for display error and drift).	T1B09
Technician power limit on HF segments	**200 watts PEP**.	T1B11
Technician power limit above 30 MHz	**1500 watts PEP** (with a few specific exceptions).	T1B12

Technicians get a slice of one HF band – 10 meters – plus all of the amateur VHF and UHF bands above it. The maps below show the bands you will use most.



The VHF/UHF bands a Technician uses most. 52.525 MHz is in 6 m; 146.52 MHz in 2 m.



The 10-meter band runs 28.0-29.7 MHz. Technicians get the bottom slice, up to 28.500: CW/digital, then phone.

Remember: a Technician’s HF voice privileges are **10 meters only**, at up to **200 watts**. Above 30 MHz the limit jumps to **1500 watts**. Digital modes are open on 10, 6, and 2 meters. And the rule behind half of these questions is simple – the FCC made amateur radio to help people **build real radio skill**, and it expects you to operate without interfering with anyone else.

Dessert 1 Question Bank (T1A + T1B)

T1A01. Which of the following is part of the Basis and Purpose of the Amateur Radio Service?
A. Providing personal radio communications for as many citizens as possible
B. Providing communications for international contesting
C. Advancing skills in the technical and communication phases of the radio art
D. All these choices are correct

T1A02. Which agency regulates and enforces the rules for the Amateur Radio Service in the United States?
A. ARRL
B. Homeland Security
C. The FCC
D. All these choices are correct

T1A03. What do the FCC rules state regarding the use of a phonetic alphabet for station identification in the Amateur Radio Service?
A. It is required when transmitting emergency messages
B. It is encouraged when using phone emissions
C. It is required when in contact with foreign stations
D. All these choices are correct

T1A04. How do you receive official notification of your new license and call sign after passing the exam?
A. Email from the FCC with a link to download the license grant
B. USPS First-Class Mail from the FCC
C. Email from the lead volunteer examiner
D. USPS First-Class Mail from the lead volunteer examiner

T1A05. What proves that the FCC has issued an operator/primary license grant?
A. A printed copy of the certificate of successful completion of examination
B. An email notification from the NCVEC granting the license
C. The license appears in the FCC ULS database
D. All these choices are correct

T1A06. On which of the following HF frequencies can automatically controlled amateur propagation beacons be found?
A. Anywhere in the Technician portion of 10 meters
B. Anywhere, if they are registered with the FCC
C. In a specific portion of each of the HF Technician bands
D. On ten meters, between 28.200 MHz and 28.300 MHz

T1A07. What is the FCC Part 97 definition of a space station?
A. Any satellite orbiting Earth
B. A manned satellite orbiting Earth
C. An amateur station located more than 50 km above Earth’s surface
D. An amateur station using amateur radio satellites for relay of signals

T1A08. Who recommends transmit/receive channels for repeater and auxiliary stations?
A. A Frequency Spectrum Manager appointed by the FCC
B. A Volunteer Frequency Coordinator recognized by local amateurs
C. An FCC Regional Field Office
D. The International Telecommunication Union

T1A09. Who selects a Frequency Coordinator?
A. The FCC Office of Spectrum Management and Coordination Policy
B. The local chapter of the Office of National Council of Independent Frequency Coordinators
C. Amateur operators in a local or regional area whose stations are eligible to be repeater or auxiliary stations
D. The FCC Regional Field Office

T1A10. Besides an FCC-issued amateur operator license, what is required to be the control operator of a Radio Amateur Civil Emergency Service (RACES) Station?
A. A written recommendation by the local ARRL Emergency Coordinator
B. Membership in the Amateur Radio Emergency Service (ARES)
C. Certification of current enrollment by a civil defense organization
D. Nothing

T1A11. Which of the following is prohibited?
A. International communications on VHF bands and higher frequencies
B. Willful or malicious interference
C. Third-party traffic using digital modes
D. All these choices are correct

T1B01. Which of the following frequency ranges are available for phone operation by Technician licensees?
A. 28.050 MHz to 28.150 MHz
B. 28.100 MHz to 28.300 MHz
C. 28.300 MHz to 28.500 MHz
D. 28.500 MHz to 28.600 MHz

T1B02. Which of the following U.S. amateur radio operators are allowed to contact the International Space Station (ISS) on VHF bands?
A. Only amateurs with a General class or higher license
B. Any amateur with a Technician class or higher license
C. Only amateurs with a General class or higher license, and NASA approval
D. Any amateurs with a Technician class or higher license, and NASA approval

T1B03. Which frequency is in the 6-meter amateur band?
A. 49.00 MHz
B. 52.525 MHz
C. 28.50 MHz
D. 222.15 MHz

T1B04. Which amateur band includes 146.52 MHz?
A. 6 meters
B. 20 meters
C. 70 centimeters
D. 2 meters

T1B05. Which of the following bands include frequencies where Technicians are authorized to use digital modes such as FT8?
A. 10 meters
B. 6 meters
C. 2 meters
D. All these choices are correct

T1B06. On which HF bands does a Technician class operator have phone privileges?
A. None
B. 10-meter band only
C. 80-meter, 40-meter, 15-meter, and 10-meter bands
D. 30-meter band only

T1B07. Which of the following VHF/UHF band segments are limited to CW only?
A. 50.0 MHz to 50.1 MHz and 144.0 MHz to 144.1 MHz
B. 219 MHz to 220 MHz and 420.0 MHz to 420.1 MHz
C. 902.0 MHz to 902.1 MHz
D. All these choices are correct

T1B08. How are US amateurs restricted in segments of bands where the Amateur Radio Service is secondary?
A. U.S. amateurs may find non-amateur stations in those segments, and must avoid interfering with them
B. U.S. amateurs must give foreign amateur stations priority in those segments
C. International communications are not permitted in those segments
D. Digital transmissions are not permitted in those segments

T1B09. Why should you not set your transmit frequency to be exactly at the edge of an amateur band or sub-band?
A. To allow for calibration error in the transmitter frequency display
B. So that modulation sidebands do not extend beyond the band edge
C. To allow for transmitter frequency drift
D. All these choices are correct

T1B10. Where may SSB phone be used in amateur bands above 50 MHz?
A. Only in sub-bands allocated to General class or higher licensees
B. Only on repeaters
C. In at least some segment of all these bands
D. On any band if the power is limited to 25 watts

T1B11. What is the maximum peak envelope power output for Technician class operators in their HF band segments?
A. 200 watts
B. 100 watts
C. 50 watts
D. 10 watts

T1B12. Except for some specific restrictions, what is the maximum peak envelope power output for Technician class operators using frequencies above 30 MHz?
A. 50 watts
B. 100 watts
C. 500 watts
D. 1500 watts

Answer Key

T1A01: C	T1A02: C	T1A03: B	T1A04: A	T1A05: C
T1A06: D	T1A07: C	T1A08: B	T1A09: C	T1A10: C
T1A11: B	T1B01: C	T1B02: B	T1B03: B	T1B04: D
T1B05: D	T1B06: B	T1B07: A	T1B08: A	T1B09: D
T1B10: C	T1B11: A	T1B12: D		